Experiment Methodology

Similarity judgement: an online two-alternative forced-choice experiment

Amit Gur, University of Antwerp · Ivana Bianchi, University of Macerata · October 2026

This document shows every page a participant sees, in the order in which the pages appear. The experiment runs in the participant's own web browser and takes 1 to 3 minutes. Pages 1 to 5 are shown once; page 6 is repeated for each of the 19 questions.

Pages

1. Welcome page
2. Participant information and consent
3. Age and gender
4. Screen calibration
5. Instructions
6. Similarity judgement question (example)
7. End

1. Welcome page

This is a study in visual perception. You will see a series of images, each featuring a reference object (**A**) at the top and two comparison objects (**B** and **C**) below it. Your task is to decide which of the two comparison objects is most similar to the reference object. This is a perception task, not an intelligence test, and your answers will help us understand which stimuli appear more similar to human beings than others. There are no right or wrong answers, so simply rely on your first visual impression.

The data will be treated anonymously.

Before you start, if you agree to participate, you will be asked to give your consent and go through a quick calibration procedure to ensure that you see the images on your screen at a standard size.

The experiment will only take 1–3 minutes.

Before we begin, please make up an 8-digit number as your participant code.

Participant code 8 DIGITS

Continue

Questions about the experiment can be sent to Amit.Gur@UAntwerpen.be

2. Participant information and consent

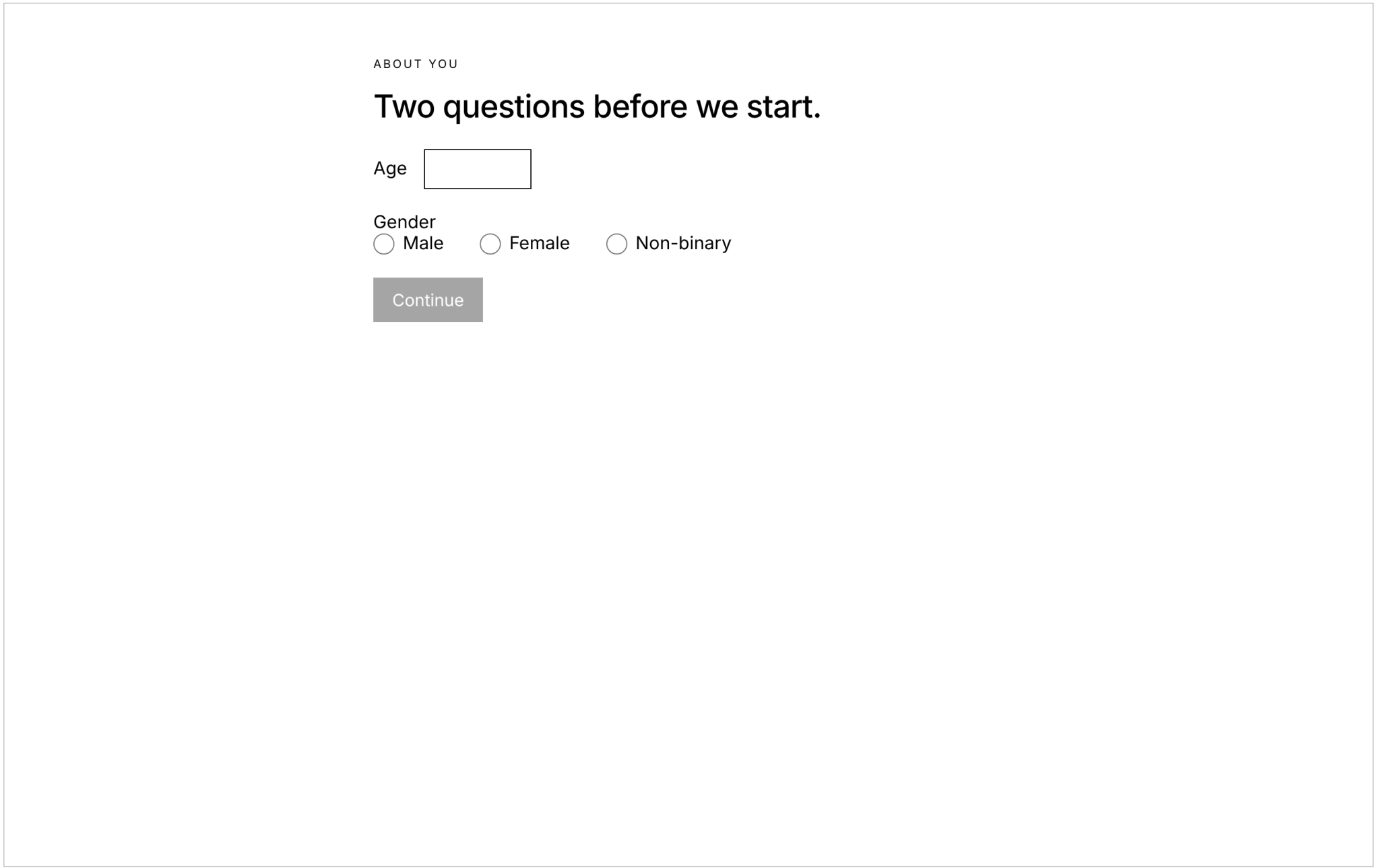
Participant Information & Consent

You are invited to take part in a scientific study on visual perception, conducted by Amit Gur and Ivana Bianchi, University of Antwerp and University of Macerata. Please read the following before deciding to participate.

- **What you will do.** You will see a series of 19 screens, each showing a reference object (A) and two comparison objects (B and C). On each screen, you decide which of the two comparison objects is most similar to the reference, by pressing the left or right arrow key or by clicking on the object. There are no right or wrong answers; just follow your visual impression. Before the task, you will be asked for your age and gender, and to match an outline on the screen to a bank card so that the objects are shown at their true size. The whole experiment takes 1 to 3 minutes.
- **Voluntary participation.** Taking part is entirely voluntary. You may withdraw at any moment by closing this window.
- **Use of data.** The data obtained in this session will be used for a scientific study and may be published in anonymized, aggregated form in scientific journals and presentations.
- **Anonymity.** No identifying information is collected. Your responses, your age and gender, and technical details of your screen are stored under an anonymous participant code only, which is removed from the dataset before publication.
- **Risks.** There are no known risks beyond those of ordinary computer use.
- **Ethical approval.** This study has been approved by the Ethics Committee for the Social Sciences and Humanities of the University of Antwerp (PENDING APPROVAL – reference number and date will be added later).
- **Contact.** Questions may be addressed to Amit.Gur@UAntwerpen.be.

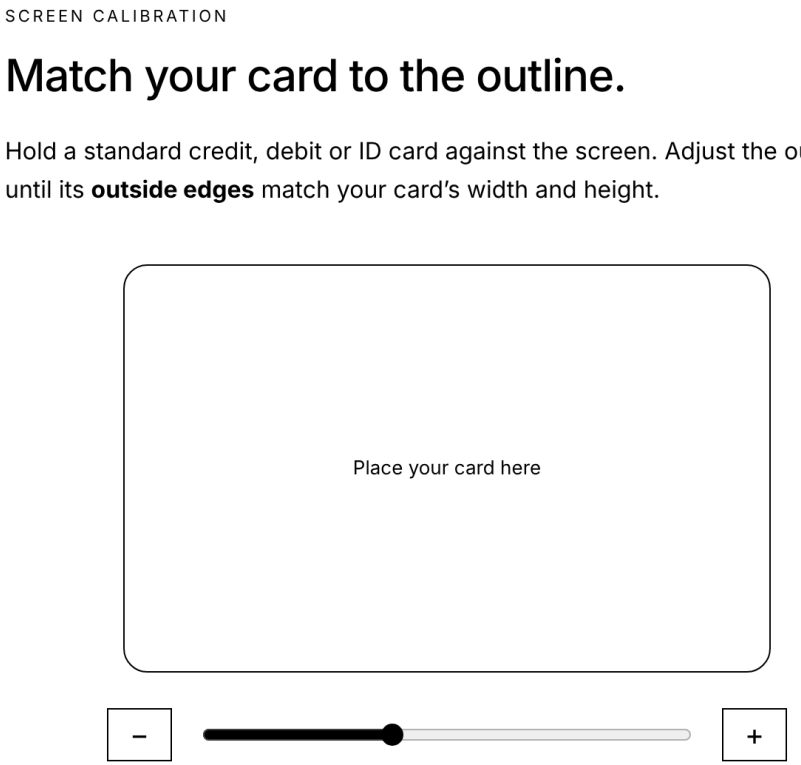
☐ I confirm that I have read and understood the information above, and that I consent to participate in this study.

3. Age and gender



4. Screen calibration

The participant holds a bank card against the screen and adjusts the outline until it matches the card. This tells the program the size of a pixel on that screen, so that the objects in the experiment are shown at their true physical size. Nothing about the card is recorded.



5. Instructions

INSTRUCTIONS

You will see a series of 19 screens, each featuring a reference object (**A**) and two comparison objects (**B** and **C**). Your task is to decide which of the comparison objects is most similar to the reference.

To choose B, press the ← LEFT arrow; to choose C, press the → RIGHT arrow. Alternatively, you can use the mouse to click on the comparison object.

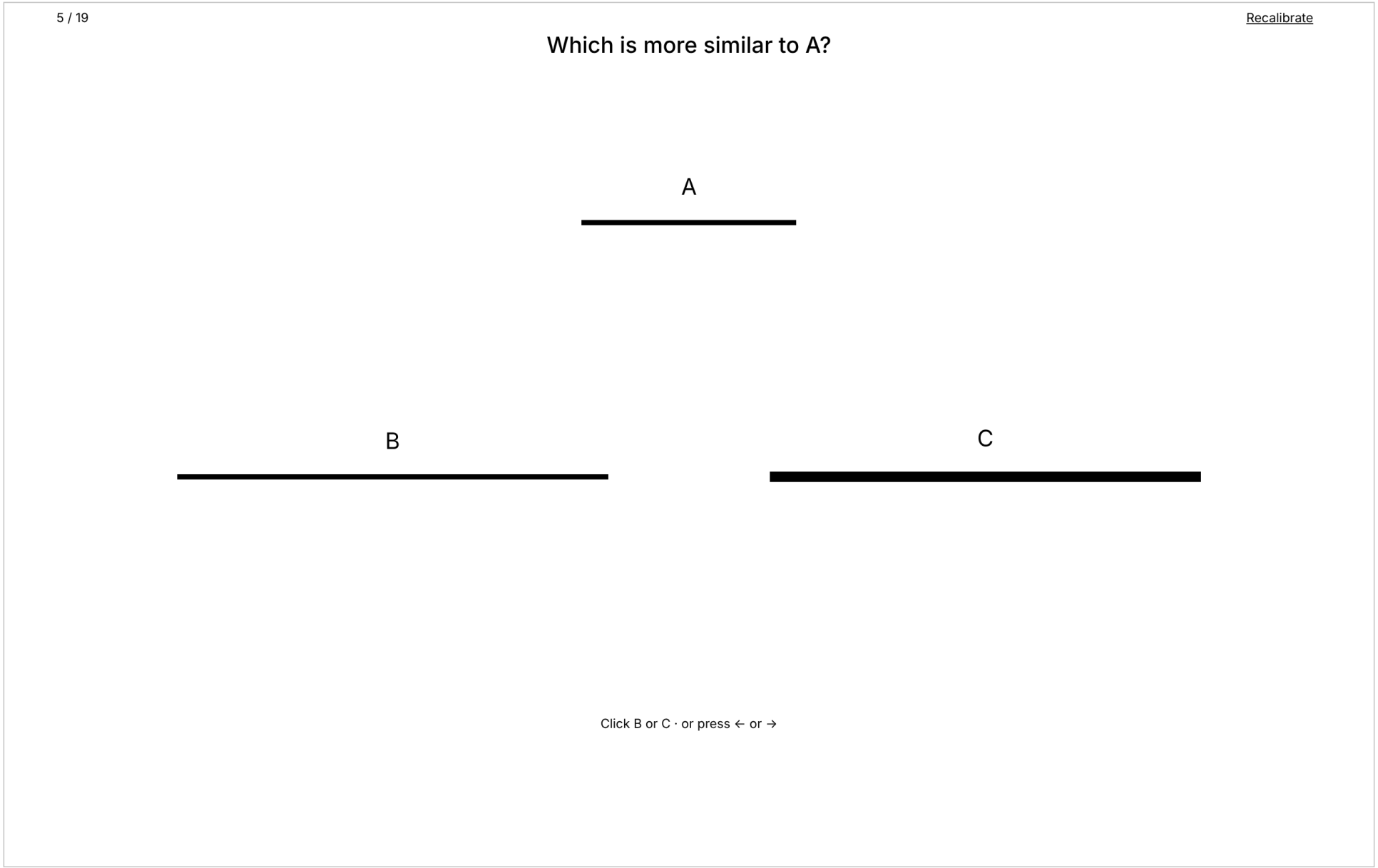
There are no right or wrong answers — just follow your visual impression.

The experiment lasts 1 to 3 minutes.

Start | Full screen | Adjust calibration

6. Similarity judgement question (example)

One of the 19 screens, which are presented in random order. The participant chooses B or C by clicking it or by pressing the left or right arrow key. There is no time limit and no feedback.



7. End

FINISHED

Thank you.